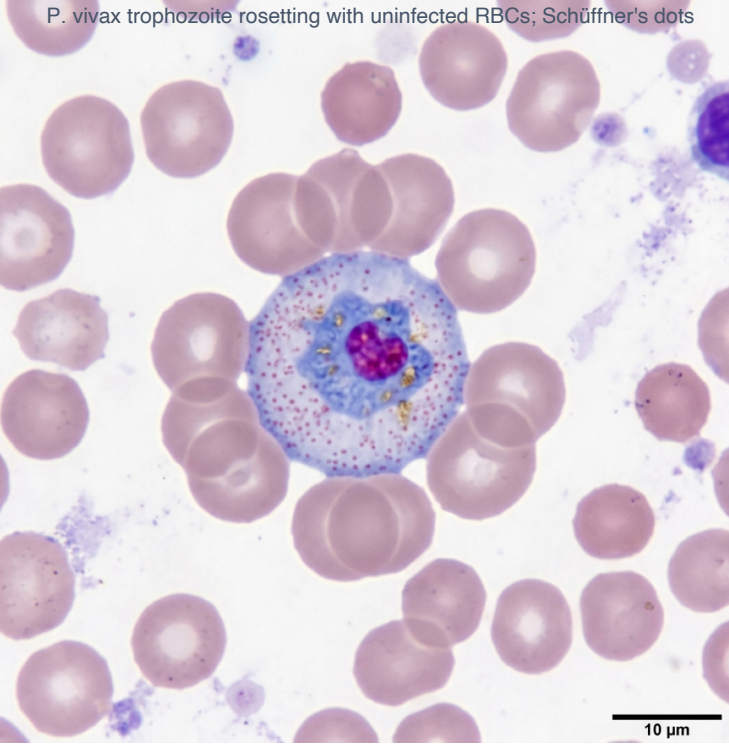
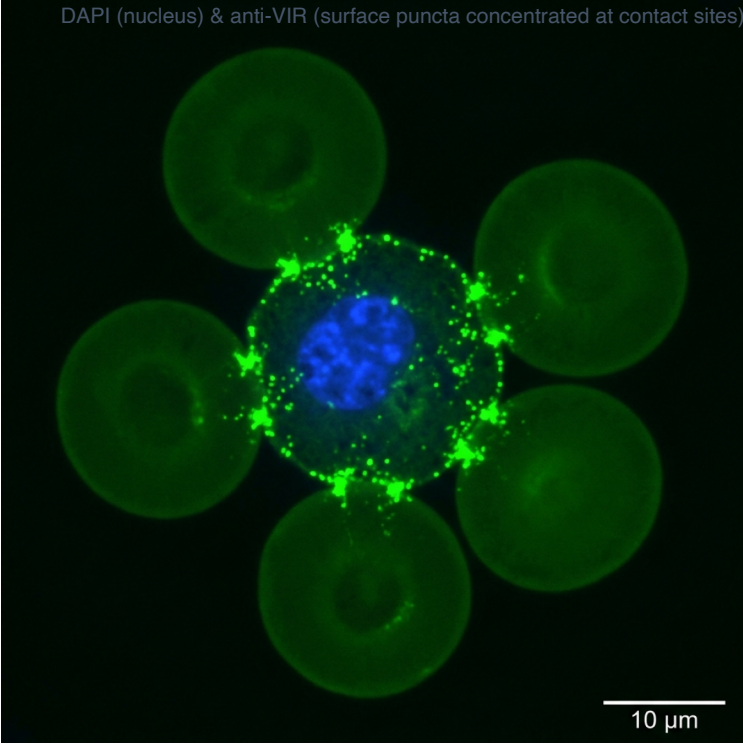


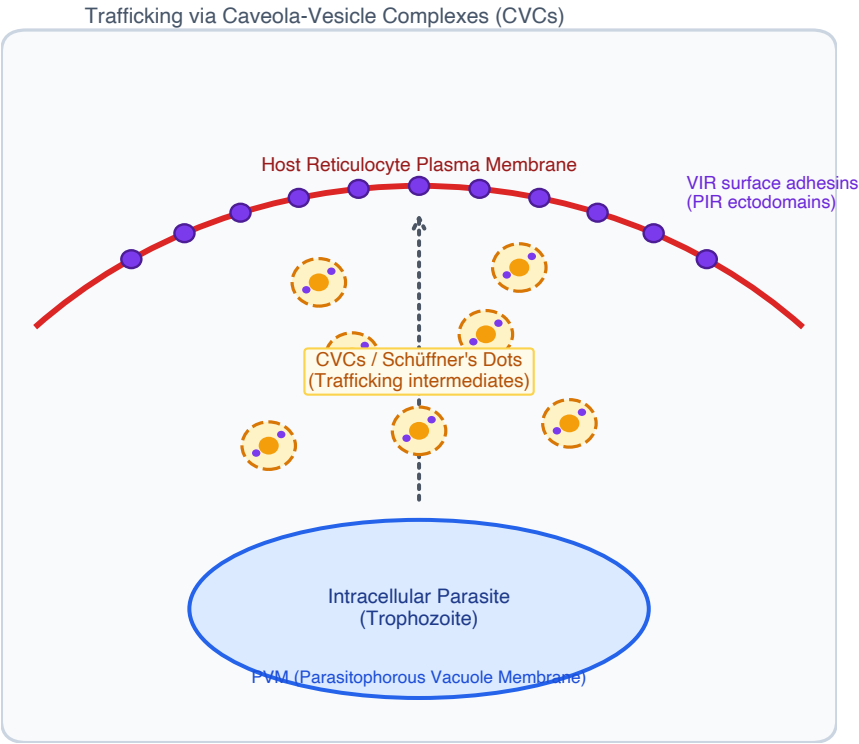
a Giemsa-Stained Light Microscopy



b Dual-Channel Confocal IFA



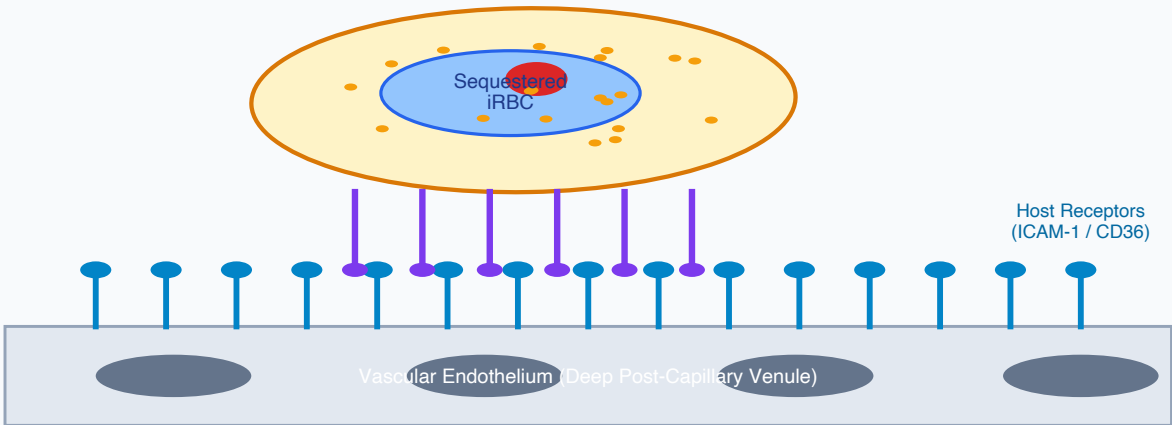
c VIR Secretory & Export Pathway



d Microvascular Cytoadherence & Immune Sequestration

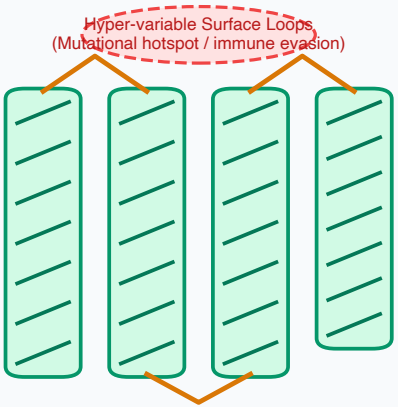
Binding endothelial receptors (ICAM-1, CD36) prevents splenic mechanical clearance

- Cytoadherence prevents splenic filtration
- Rosetting causes microvascular hypoperfusion
- Clinical severity: severe anemia & acute lung injury (ARDS)



e Conserved PIR Ectodomain Fold & 3Di Structural Alphabet

PDB 6ZYV: rigid helical bundle bridges the sequence twilight zone



Structural Homology vs Sequence Twilight Zone:

- Sequence Identity: Median ~32% within FS_001 (indistinguishable from sequence twilight noise)
- ProstT5 Transformer: Predicts 20-state 3Di backbone conformation directly from amino acid FASTA
- Foldseek Clustering: Unifies 146 sequence subfamilies into 73 structural classes
- Dominant Fold (FS_001): Encompasses 46% of all VIRs across Indonesia, Thailand, and Peru

Sequence Baseline (MCL)
146 subfamilies (over-split)

Structural Fold (Foldseek)
73 classes (50% reduction)